

Quick Reference for Production Game

Factory

Cobb-Douglas Production Function (yearly throughput): $Y = AK^{\alpha}L^{\beta}$

Cobb-Douglas Production Function (daily throughput): $\lambda = \frac{AK^{\alpha}(L*364)^{\beta}}{364}$

Little's Law: $INV = \lambda * CT$

Starting factory type: Bench

Bench factory setup time: 0.05 days; **A:** 0.009; **α :** 0.10; **β :** 0.85

Starting factory investment: \$100,000 capital + \$100,000 land

Starting daily expenditure: \$2,500

Add-on capex: 30 days

Capital depreciation: 15 years

Distribution Center

Investment: \$2,500,000 capital + \$100,000 land

Capital Depreciation: 15 years

Daily Expenditure: \$2,000

Handling for retail sales: \$10/unit

Commission for retail sales: 20% of revenue

Shipping from Factory to DC *(paid by the shipper)*

Container in region: \$5,000 per 1,000 units; 7 days

Container between region: \$10,000 per 1,000 units; 21 days

Mail in region: \$200 per 10 units; 1 day

Mail between regions: \$400 per 10 units; 3 days

Shipping from DC to DC *(paid by the shipper)*

Container in region: \$5,000 per 1,000 units; 7 days

Container between region: \$10,000 per 1,000 units; 21 days

Mail in region: \$200 per 10 units; 1 day

Mail between regions: \$400 per 10 units; 3 days

Focus Group

Size: 10 participants

Cost: \$20,000

Time: 7 days

Financial

Asset cost of capital: 15% APR

Emergency Loan: 40% APR

Interest earned on cash balances: 3% APR

Dividends: 6.5% APR after-tax return on any dividends that are paid out to shareholders

Return to Investors: cash on hand - debt + dividends paid + after tax return on dividends distributed

Trade credit terms for raw materials payable: 30 days to pay; **other payables:** 15 days to pay

Tax rate: 35%

Taxes Payable accumulate daily, and the account balance is paid on the last day of each quarter